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Experiment 1: Determination of Soil pH by Using pH Meter

Objective

To determine the active acidity or alkalinity (pH) of a given soil sample in order to assess its suitability for specific crops and to determine the necessary soil amendments.

Principle

pH is technically defined as the negative base-10 logarithm of the hydrogen ion concentration expressed in gram equivalents per litre. The pH of soil water is a crucial parameter that indicates the soil's acidity or alkalinity. It is measured on a scale of 0 to 14, where a pH of 7 is neutral, values less than 7 are acidic, and values greater than 7 are alkaline. Monitoring soil pH is essential for assessing its suitability for various biological and agronomic processes.

$$pH = -\log_{10}[H^+]$$

where $[H^+]$ is the hydrogen ion concentration. The hydrogen ion concentration is related to the dissociation constant of water as follows:

$$H_2O \rightarrow H^+ + OH^-$$
$$K = [H^+][OH^-] = 10^{-14}$$

A neutral solution has a hydrogen ion concentration of 10^{-7}, i.e. pH = 7.

Apparatus

- Air-dried soil sample
- Beaker, 100 mL
- Glass electrode pH meter (Model: HI700663)
- Stirring rod
- Distilled water
- Weighing balance
- Measuring cylinder

Procedure

1. The soil sample was collected from the laboratory.
2. The sample was ground and passed through a 2 mm sieve.
3. 10 g of soil was weighed into a 100 mL beaker using a weighing balance, and 25 mL of distilled water was added.
4. The suspension was stirred several times over the next 30 minutes.
5. The pH meter was switched on and allowed to stabilise.
6. The electrode was placed into the soil suspension, and the reading was recorded once the display had stabilised.

Observation

The stabilised reading obtained from the pH meter for the given soil sample was:

pH = 6.64

Result

Result: The pH of the given soil sample was found to be 6.64. This value falls in the slightly acidic range. At this pH, most essential plant nutrients remain reasonably available, although prolonged cultivation without liming may gradually lower the pH further and reduce the availability of nutrients such as phosphorus and molybdenum.

Discussion

On commonly used soil-reaction classification scales, a pH of 6.64 falls at the boundary between the "slightly acidic" (6.1–6.5) and "neutral" (6.6–7.3) classes, indicating a soil reaction close to the optimum range for most agronomic crops. This is broadly consistent with the small lime requirement of only 2.5 g $CaCO_3$ per kg of soil obtained in Experiment 3, since a soil already close to neutrality needs only a modest addition of base to reach the target pH. As this determination relies on a single stabilised meter reading rather than a titration, the main sources of error are incomplete equilibration of the soil–water suspension (mitigated here by the 30-minute stirring period) and drift in electrode calibration, which should be checked periodically with standard buffer solutions.

Experiment 2: Determination of the Electrical Conductivity (EC) of Soil

Objective

To determine the electrical conductivity of a given soil sample in order to estimate the concentration of total soluble salts and assess the potential salinity hazard for crop production.

Principle

Electrical conductivity (EC) is a measure of a solution's ability to conduct an electric current, which depends on the concentration of dissolved ions such as salts and minerals. It is typically expressed in microsiemens per centimetre (µS/cm). Pure water is a very poor conductor of electricity, whereas water containing dissolved salts conducts current approximately in proportion to the amount of salt present. Electrical conductance, the reciprocal of resistance, is therefore used for salinity appraisal.

For example: EC = 0.000521 mhos/cm = 0.521 millimhos/cm (mS/cm) = 521 micromhos/cm (µS/cm).

Apparatus

- Air-dried soil sample
- Beaker, 100 mL
- Glass electrode EC meter (JENCO – EC330)
- Stirring rod
- Distilled water
- Weighing balance
- Measuring cylinder

Procedure

1. 10 g of soil was weighed using a weighing balance.
2. The soil sample was poured into a 100 mL beaker.
3. 50 mL of distilled water was added to the sample.
4. The suspension was stirred for 30 minutes with a glass rod.
5. After 30 minutes, the EC meter electrode was placed in the beaker.
6. Once the reading had stabilised, the EC value was recorded.

Observation

The stabilised reading obtained from the EC meter for the given soil sample was:

EC = 125 µS/cm

Result

Result: The EC value of the soil sample was found to be 125 µS/cm. This indicates a low soluble-salt content, which is generally favourable for plant growth. The soil is non-saline and is unlikely to cause any salinity stress to most crops.

Discussion

Expressed in the more commonly cited units of deciSiemens per metre, 125 µS/cm corresponds to about 0.125 dS/m, well below the threshold of roughly 2 dS/m generally used to define the lower boundary of “slightly saline” soils. The soil can therefore be classified as non-saline, and salinity is not expected to limit crop establishment or yield. As with the pH determination, this result is based on a single stabilised meter reading; the main precautions needed to ensure accuracy are thorough mixing of the soil–water suspension for the full recommended period and periodic calibration of the EC probe against a standard KCl solution, since electrode fouling or drift can otherwise bias the reading.

Experiment 3: Lime Requirement of Soil Sample

Objective

To determine the amount of agricultural lime required to neutralise soil acidity and raise the pH of a given soil sample to a target optimum level for crop production.

Principle

Plant growth is adversely affected by low soil pH. To bring the pH back towards neutrality, lime is added to the soil. Liming may be carried out using CaO, $Ca(OH)_2$, $CaCO_3$ (calcite), or $CaCO_3 \cdot MgCO_3$ (dolomite). Soil pH is the most important criterion determining the lime requirement, although texture, structure, organic-matter content, and the type and amount of clay also influence it. Lime neutralises soil acidity by reacting with H^+ ions in the soil solution, forming water and carbon dioxide and thereby raising the pH:

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

Apparatus

- Beaker, 50 mL
- Digital pH meter (LAQUA model)
- Dropper
- Glass stirring rod

Reagent

Calcium hydroxide, $Ca(OH)_2$ solution (0.01 N).

Procedure

1. 10 g of soil was taken, 25 mL of distilled water was added, and the mixture was shaken for 30 minutes.
2. The pH of the resulting soil suspension was measured using the pH meter; this was recorded as the '0 drop' reading.
3. One drop of $Ca(OH)_2$ solution was added, the suspension was mixed, and the pH was recorded.
4. The addition of a single drop of $Ca(OH)_2$ was repeated, with the pH recorded after each individual drop, up to a total of ten drops.

Data Table

Lime drop, $Ca(OH)_2$	pH
0	6.94
1	7.04
2	7.13
3	7.32

Lime drop, $Ca(OH)_2$	pH
4	7.48
5	7.58
6	7.64
7	7.67
8	8.04
9	8.13
10	8.19

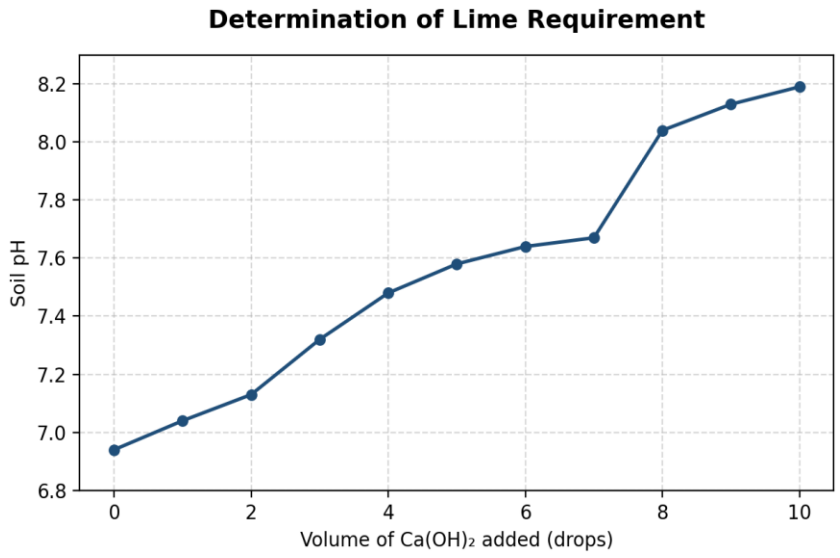


Fig. 3.1: Change in soil pH with successive additions of $Ca(OH)_2$ (Determination of Lime Requirement)

Calculation

One drop of $Ca(OH)_2$ is taken as ≈ 0.05 mL (standard drop volume).

$$\textit{Total volume of } Ca(OH)_2 \textit{ added (10 drops)} = 10 \times 0.05 = 0.5 \textit{ mL}$$

Since 10 g of soil was used, the scaling factor to express the result per kg of soil is:

$$\textit{Scaling factor} = 1000 / 10 = 100$$

The lime requirement is calculated using the relation:

$$\textbf{\textit{Lime requirement (g } } CaCO_3 \textbf{\textit{ per kg soil)}} = [\,(V \times N \times 50) / \textit{Weight of soil (g)}\,] \times \textit{Scaling factor}$$

where V is the total volume of $Ca(OH)_2$ added, N is the normality of the $Ca(OH)_2$ solution, and 50 is the equivalent-weight conversion factor for $CaCO_3$. Here, V = 0.5 mL, N = 0.01 N, weight of soil = 10 g, and scaling factor = 100. Substituting these values:

Lime requirement = [(0.5 × 0.01 × 50) / 10] × 100

= 2.5 g $CaCO_3$ per kg of soil

Result

Result: The lime requirement of the given soil sample was found to be 2.5 g of $CaCO_3$ per kg of soil. The titration curve shows that soil pH rose steadily from 6.94 to 8.19 with successive additions of $Ca(OH)_2$, with a comparatively sharper rise beyond the 7th drop, indicating the buffering capacity of the soil against the applied base.

Discussion

The pH-versus-lime curve rises gradually up to about the 7th drop and then more steeply thereafter, reflecting the soil's buffering capacity: the initial additions of base are consumed in neutralising readily exchangeable acidity, after which further small additions of $Ca(OH)_2$ produce a comparatively larger rise in pH. Because the target pH for most crops is close to neutral (pH 6.5–7.0), and the soil's initial pH (6.94) was already close to this range, only a modest lime application (2.5 g $CaCO_3$/kg soil) was calculated as necessary. The main sources of error in this method are the assumption of a fixed drop volume (0.05 mL), which can vary slightly with dropper technique and solution viscosity, and any drift in the pH electrode calibration during the course of the titration; both can be minimised by periodically recalibrating the meter and dispensing the reagent from a calibrated micro-burette rather than a dropper where higher precision is required.

Experiment 4: Determination of Organic Carbon of Soil Sample

(Walkley and Black Wet Oxidation Method)

Objective

To determine the percentage of organic carbon present in a soil sample by the Walkley and Black wet-oxidation method.

Principle

Soil organic carbon is oxidised by potassium dichromate ($K_2Cr_2O_7$) in the presence of concentrated sulfuric acid (H_2SO_4); the heat generated on dilution of the acid assists the oxidation of organic matter. Organic-carbon compounds act as strong reducing agents, and this reducing capacity is exploited to determine organic carbon by digestion with a standard oxidising agent — chromic acid. The reaction may be represented as:

$$4Cr^{6+} + 3C \rightarrow 4Cr^{3+} + 3C^{4+}$$

$$K_2Cr_2O_7 + 3C + 6H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 3CO_2 + 8H_2O + H_2Cr_2O_7\ (unreduced)$$

The excess (unreduced) chromic acid remaining after oxidation of the organic carbon is back-titrated with a standard ferrous sulfate solution:

$$K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$$

and the amount of carbon oxidised is calculated from the quantity of chromic acid reduced.

Apparatus

- Burette
- 10 mL pipette
- 500 mL conical flask
- 10 mL and 100 mL measuring cylinders
- Distilled water bottle
- Analytical balance
- Retort stand and clamp
- Heat-resistant sheet

Reagents

- Concentrated sulfuric acid (≥ 96%)
- Concentrated phosphoric acid (85%)
- Sodium fluoride (NaF)
- Standard 1 N potassium dichromate solution (49.03 g $K_2Cr_2O_7$ dissolved and diluted to 1 L)
- Ferrous sulfate solution (278 g $FeSO_4 \cdot 7H_2O$ + 15 mL conc. H_2SO_4, diluted to 1000 mL)
- Diphenylamine indicator

Procedure

1. 2 g of air-dried soil (passed through a 2 mm sieve) was transferred to a 500 mL clean, dry conical flask.
2. 10 mL of standard 1 N $K_2Cr_2O_7$ solution was added using a pipette.
3. 10 mL of concentrated H_2SO_4 was added with a measuring cylinder and mixed thoroughly.
4. The flask was allowed to cool on a heat-resistant sheet with occasional shaking for 30 minutes.
5. After cooling, about 150 mL of distilled water, 10 mL of phosphoric acid, and 0.2 g of sodium fluoride were added.
6. 3 mL (about 60 drops) of diphenylamine indicator was added, producing a deep violet colour.
7. The excess (unreduced) chromic acid was titrated with standard ferrous sulfate solution from a burette until the colour changed sharply to deep bottle-green (the end point).
8. The titration was repeated three times, and a blank was run in the same manner, omitting the soil, to standardise the $FeSO_4$ solution.

Data Table

	No. of observation	Initial burette reading (mL)	Final burette reading (mL)	Difference (mL)	Mean (mL)
Soil sample (T)	1	10	25	15.0	14.65
	2	25	39.3	14.3	
Blank (B)	1	0	20.6	20.6	20.45
	2	20.6	40.9	20.3	

Standardisation of $FeSO_4$ Solution

Using $S_1V_1 = S_2V_2$, where S_1 = normality of $K_2Cr_2O_7$ = 1 N, V_1 = 10 mL, V_2 = mean volume of $FeSO_4$ used in the blank = 20.45 mL:

$$S_2 = (S_1 \times V_1) / V_2$$
$$= (1 \times 10) / 20.45$$
$$= 0.489 \text{ N}$$

Calculation

Since 1000 mL of 1 N $K_2Cr_2O_7$ is equivalent to 3 g of carbon (equivalent weight of C = 12/4 = 3), 1 mL of 1 N $K_2Cr_2O_7$ corresponds to 0.003 g of C. As only about 77% of soil carbon is oxidised by 1 N $K_2Cr_2O_7$ under these conditions, the factor 0.003 is multiplied by 1.3 to obtain total organic carbon:

$$\% \text{ Organic carbon} = [(B - T) \times f \times 0.003 \times 1.3 \times 100] / W$$

where B = mean mL of $FeSO_4$ for the blank = 20.45, T = mean mL of $FeSO_4$ for the soil sample = 14.65, f = strength of $FeSO_4$ (S_2) = 0.489, and W = weight of soil taken = 2 g.

$$\% \text{ Organic carbon} = [(20.45 - 14.65) \times 0.489 \times 0.003 \times 1.3 \times 100] / 2 = 0.55\%$$

Result

Result: The organic carbon content of the given soil sample was found to be 0.55%. This means that every 100 g of soil contains approximately 0.55 g of organic carbon, indicating a moderate level of organic matter in the soil.

Discussion

Multiplying the organic carbon percentage by the conventional Van Bemmelen factor of 1.724 (which assumes organic matter contains, on average, 58% carbon) gives an estimated soil organic matter content of about 0.55 × 1.724 ≈ 0.95%, which is on the lower side for agricultural soils and suggests that the field would benefit from organic amendments such as compost or crop-residue incorporation. The duplicate titre readings for the soil sample (15.0 and 14.3 mL) and for the blank (20.6 and 20.3 mL) are reasonably close, indicating acceptable titration precision. Because the Walkley–Black method oxidises only about 77% of the total soil carbon (hence the 1.3 correction factor applied above), the result should be regarded as an estimate rather than an absolute value; incomplete mixing of the $K_2Cr_2O_7$–H_2SO_4 digestion mixture and subjectivity in judging the diphenylamine end-point colour change are the main sources of error, and can be reduced by ensuring vigorous initial mixing and titrating slowly near the end point.

Experiment 5: Determination of Total Nitrogen in Soil Sample

(Kjeldahl Method)

Objective

To determine the percentage of total nitrogen present in a given soil sample by the Kjeldahl method.

Principle

The Kjeldahl procedure for the determination of total nitrogen involves three steps. First, digestion: organic nitrogen in the soil is converted into ammonium sulfate by heating with concentrated sulfuric acid in the presence of a catalyst. Second, distillation: the digested solution is made alkaline with sodium hydroxide, which liberates ammonia gas. Third, titration: the liberated ammonia is absorbed in boric acid and titrated against a standard acid.

$$\text{Soil–N} + H_2SO_4 \xrightarrow{\text{(heat, 360–410°C)}} (NH_4)_2SO_4$$

$$(NH_4)_2SO_4 + 2NaOH \rightarrow NH_3\uparrow + H_2O + Na_2SO_4$$

$$NH_3 + H_3BO_3 \rightarrow NH_4H_2BO_3 \text{ (ammonium borate)}$$

$$NH_4H_2BO_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4 + H_3BO_3$$

The digestion temperature is critical: below 360°C the release of ammonia is slow and incomplete, while above 410°C some ammonia is lost from the mixture.

Apparatus

- Kjeldahl digestion unit
- 500 mL Kjeldahl flask
- Distillation apparatus
- 25 mL conical flask
- 10 mL and 50 mL measuring cylinders
- 10 mL pipette
- Micro-burette
- Retort stand

Reagents

- Concentrated sulfuric acid
- Digestion mixture ($CuSO_4 \cdot 5H_2O$, selenium powder, and anhydrous Na_2SO_4)
- 40% NaOH solution
- Mixed indicator (bromocresol green and methyl red in 95% ethanol)
- 4% boric acid – mixed indicator solution
- Standard N/56 sulfuric or hydrochloric acid

Procedure

1. 5 g of finely powdered soil was placed in a 500 mL Kjeldahl flask, 20 mL of distilled water was added, and the mixture was left for 20 minutes.
2. 25 mL of concentrated H_2SO_4 was added and mixed thoroughly, and the flask was heated over a low flame for 15 minutes.
3. When white fumes of H_2SO_4 appeared, the flask was removed and 2–3 g of catalyst mixture was added.
4. Digestion was continued with slowly rising temperature for two hours or more until the liquid became clear (light bluish colour).
5. The digest was cooled, diluted with 100 mL distilled water, and transferred to a 250 mL volumetric flask made up to volume.
6. An aliquot of the extract was distilled with 40% NaOH using the micro-Kjeldahl distillation apparatus, and the liberated ammonia was collected in 10 mL of 4% boric acid containing mixed indicator.
7. The distillate was titrated against standard H_2SO_4 until the colour changed to pink, marking the end point.
8. A blank determination was carried out simultaneously using all reagents but no soil.

Data Table

	No. of observation	Initial burette reading (mL)	Final burette reading (mL)	Difference (mL)	Average (mL)
Soil sample (T)	1	1.1	3.1	2.0	1.8
	2	3.1	4.7	1.6	
Blank (B)	1	6.7	8.2	1.5	1.2
	2	8.2	9.1	0.9	

T = 1.8 mL, B = 1.2 mL, f (normality factor of standard acid) = 0.018, w (weight of soil) = 5 g, volume of extract used = 10 mL.

Calculation

Since 1000 mL of 1 N H_2SO_4 is equivalent to 14 g of N, 1 mL of 1 N H_2SO_4 corresponds to 0.014 g of N. The percentage of nitrogen in the soil is calculated as:

$$\% \text{ N in soil} = [(T - B) \times f \times 0.014 \times 250 \times 100] / (w \times \text{volume of extract used})$$

$$\% \text{ N in soil} = [(1.8 - 1.2) \times 0.018 \times 0.014 \times 250 \times 100] / (5 \times 10)$$

$$= 0.0756\%$$

Result

Result: The total nitrogen content of the given soil sample was found to be 0.0756%, equivalent to 0.756 g N per kg of soil.

Discussion

The nitrogen level obtained (0.756 g N/kg soil) falls in the low range typical of many cultivated tropical and sub-tropical soils, where continuous cropping without adequate residue return or nitrogenous fertilisation tends to deplete soil nitrogen reserves. Comparing this result with the organic carbon content obtained in Experiment 4 (0.55%) gives an approximate C : N ratio of 0.55 / 0.0756 ≈ 7.3 : 1, which is somewhat narrower than the C : N ratio of about 10–12 : 1 typically reported for surface agricultural soils; such a narrow ratio can indicate rapid mineralisation of organic matter or may simply reflect the combined analytical uncertainty of the two wet-chemistry methods, and would be worth verifying with repeat analyses. The duplicate blank titres (1.5 and 0.9 mL) show a wider spread than the soil-sample duplicates (2.0 and 1.6 mL), suggesting that the blank titration end point was harder to judge precisely; possible sources of error include loss of ammonia during distillation if the delivery tube is not properly sealed below the boric acid surface, incomplete digestion of organic nitrogen, and indicator end-point subjectivity, all of which can be reduced by ensuring a tight distillation assembly and consistent indicator volumes.

Experiment 6: Determination of Calcium (Ca^{2+}) and Magnesium (Mg^{2+}) by EDTA Titration

Objective

To determine the concentration of exchangeable Ca^{2+} and Mg^{2+} ions in a given soil sample by complexometric titration with EDTA.

Principle

Calcium and magnesium present in the soil are extracted using 1 N ammonium acetate (NH_4OAc) solution. The extracted Ca and Mg are then determined by titration with ethylenediaminetetraacetic acid (EDTA), which forms stable, soluble 1:1 complexes with both Ca^{2+} and Mg^{2+}:

$$Ca^{2+} + EDTA^{4-} \rightarrow [Ca–EDTA]^{2-}$$

$$Mg^{2+} + EDTA^{4-} \rightarrow [Mg–EDTA]^{2-}$$

For calcium alone, murexide indicator is used at about pH 11; the colour changes from pink to light blue at the end point (magnesium is masked as $Mg(OH)_2$ precipitate). For combined calcium and magnesium, Eriochrome Black-T (EBT) indicator is used at about pH 10; calcium is complexed first, after which the remaining magnesium reacts with EDTA. The magnesium content is then obtained by subtracting the calcium value from the combined (Ca + Mg) value.

Apparatus

- 250 mL conical flask
- Filter funnel
- Filter paper (Whatman No. 40/42)
- Micro-burette
- Small porcelain cup
- Measuring cylinder
- Pipette
- Volumetric flask
- Beaker

Reagents

- 1 N ammonium acetate solution (pH ≈ 7.0)
- 10% sodium hydroxide solution
- Murexide indicator (0.5 g murexide + 100 g K_2SO_4)
- Standard EDTA solution (≈ 0.005 M / 0.01 N)
- pH 10 buffer solution (NH_4Cl + NH_4OH)
- Hydroxylamine hydrochloride solution
- Potassium ferrocyanide solution

- Triethanolamine (TEA)
- Eriochrome Black-T (EBT) indicator

Procedure

A. Extraction of Ca and Mg

1. 10 g of air-dried soil was taken in a 250 mL conical flask and 100 mL of neutral 1 N NH_4OAc solution was added.
2. The flask was shaken thoroughly and allowed to stand overnight, then filtered through Whatman No. 40/42 paper to obtain the soil extract.

B. Determination of Calcium

1. 2 mL of soil extract was taken in a porcelain cup and 5 mL of 10% NaOH solution was added to precipitate $Mg(OH)_2$.
2. About 0.02 g of murexide indicator was added, and the solution was titrated with 0.01 N EDTA until the colour changed from pink to light blue (end point).
3. A blank was run using distilled water in place of soil extract, and the EDTA was standardised against a standard calcium solution.

C. Determination of Calcium + Magnesium

1. 2 mL of soil extract was taken in a porcelain cup with 25 mL distilled water, 10–15 mL of pH 10 buffer, and a few drops each of potassium ferrocyanide, hydroxylamine hydrochloride, and TEA solutions.
2. The solution was warmed gently, cooled, and 3 drops of Eriochrome Black-T indicator was added.
3. The solution was titrated with EDTA until the colour changed at the end point, and a corresponding blank was run.

Observation

Calcium determination

	No. of observation	Initial burette reading (mL)	Final burette reading (mL)	Difference (mL)	Mean (mL)
Soil sample	1	0	5.2	5.2	5.2
	2	5.2	10.4	5.2	
Blank	1	33.5	34.5	1.0	1.0

Corrected Ca reading = 5.2 − 1.0 = 4.2 mL

Calcium + Magnesium determination

	No. of observation	Initial burette reading (mL)	Final burette reading (mL)	Difference (mL)	Mean (mL)
Soil sample	1	0	8.2	8.2	8.1
	2	8.2	16.2	8.0	
Blank	1	0	1.0	1.0	1.0

Corrected (Ca + Mg) reading = 8.1 − 1.0 = 7.1 mL

Mg reading = (Ca + Mg) reading − Ca reading = 7.1 − 4.2 = 2.9 mL

Calculation

Calcium: since 1000 mL of 1 N EDTA ≡ 20 g Ca, 1 mL of 1 N EDTA ≡ 0.02 g Ca.

% Ca in soil = [0.02 × mL of EDTA used × Normality of EDTA × Total extract volume (mL) × 100] / (mL of extract used × Weight of soil)

With corrected EDTA volume = 4.2 mL, normality of EDTA = 0.0048 N, extract used = 2 mL, weight of soil = 10 g, and total extract volume = 100 mL:

% Ca = (0.02 × 4.2 × 0.0048 × 100 × 100) / (2 × 10) = 0.20% (= 2000 ppm)

Magnesium: since 1000 mL of 1 N EDTA ≡ 12 g Mg, 1 mL of 1 N EDTA ≡ 0.012 g Mg.

% Mg in soil = [0.012 × mL of EDTA used × Normality of EDTA × Total extract volume (mL) × 100] / (mL of extract used × Weight of soil)

With corrected Mg-EDTA reading = 2.9 mL, normality of EDTA = 0.0048 N, extract used = 2 mL, weight of soil = 10 g, total extract volume = 100 mL:

% Mg = (0.012 × 2.9 × 0.0048 × 100 × 100) / (2 × 10) = 0.0835% (= 835 ppm)

Result

Nutrient	Percentage (%)	Concentration (ppm)
Calcium (Ca)	0.20	2000
Magnesium (Mg)	0.0835	835

Result: The available calcium and magnesium contents of the soil sample were found to be 0.20% (2000 ppm) and 0.0835% (835 ppm), respectively, indicating that the soil has an adequate base status for most crops.

Discussion

The Ca : Mg ratio of the soil, calculated from the results above, is approximately 2000 / 835 ≈ 2.4 : 1. Agronomically, a Ca : Mg ratio in the range of about 3–6 : 1 is often considered favourable for nutrient balance, so a ratio below this range may indicate a comparatively high proportion of exchangeable magnesium, which can occasionally interfere with potassium and calcium uptake in sensitive crops. The close agreement between the duplicate titre readings in both the calcium and combined calcium-plus-magnesium titrations (5.2 and 5.2 mL; 8.2 and 8.0 mL) indicates good titration precision. Principal sources of error in this method include subjectivity in judging the murexide and Eriochrome Black-T colour end points, incomplete masking of interfering ions, and any loss of extract during filtration, all of which can be minimised by titrating slowly near the end point and using freshly prepared indicator solutions.

Experiment 7: Determination of Chloride (Cl^-) in Soil Sample

(Mohr's Argentometric Titration Method)

Objective

To determine the amount of chloride present in a soil sample by extracting soluble chloride with distilled water and titrating it against standard silver nitrate ($AgNO_3$), using potassium chromate (K_2CrO_4) as the indicator.

Principle

Chloride ions in the soil extract react with silver ions from $AgNO_3$ to form a white precipitate of silver chloride:

$$Ag^+ + Cl^- \rightarrow AgCl\downarrow$$

Once all the chloride has precipitated, the first excess of silver ions reacts with the chromate ions of the indicator to form brick-red silver chromate, marking the end point:

$$2Ag^+ + CrO_4^{2-} \rightarrow Ag_2CrO_4\downarrow \text{ (brick-red)}$$

Apparatus

- Analytical balance
- Conical flask
- Measuring cylinder
- Volumetric flask
- Pipette
- Burette
- Funnel
- Filter paper
- Mechanical shaker
- Beaker
- Wash bottle

Reagents

- 0.01 N standard $AgNO_3$ solution
- 5% K_2CrO_4 indicator solution (5 g K_2CrO_4 dissolved and made up to 100 mL)

Procedure

Preparation of soil extract

1. 10 g of air-dried, sieved soil was weighed into a 250 mL conical flask, and 50 mL of distilled water was added (soil : water = 1 : 5).

2. The mixture was shaken for 30 minutes, allowed to settle, and filtered through Whatman No. 42 filter paper to obtain the clear soil extract.

Titration

1. For the blank, 10 mL of distilled water was pipetted into a conical flask and 1 mL of 5% K_2CrO_4 indicator was added, then titrated with standard $AgNO_3$ until a permanent faint brick-red colour appeared.
2. For the sample, 10 mL of soil extract was pipetted into a clean conical flask, 1 mL of indicator was added (the solution turned yellow), and the mixture was titrated with 0.01 N $AgNO_3$ with continuous swirling until the first permanent brick-red/reddish-brown colour appeared.
3. The titration was repeated to obtain concordant readings.

Data Table

No. of observation	Initial burette reading (mL)	Final burette reading (mL)	Difference (mL)	Mean (mL)
Soil sample – 1	0	3.5	3.5	3.5
Soil sample – 2	3.5	7.0	3.5	
Blank	7.0	8.8	1.8	1.8

Calculation

Given: weight of soil = 10 g, total extract volume = 50 mL, volume of extract titrated (V_s) = 10 mL, normality of $AgNO_3$ (N) = 0.01 N, equivalent weight of Cl^- = 35.45 g/eq, sample titre (V) = 3.5 mL, blank titre (B) = 1.8 mL.

$$Cl^- \text{ (mg/L)} = [\,(V - B) \times N \times 35.45 \times 1000\,] / V_s$$

$$Cl^- \text{ (mg/L)} = [\,(3.5 - 1.8) \times 0.01 \times 35.45 \times 1000\,] / 10$$

$$= 60.265 \text{ mg/L}$$

Accounting for the soil-to-water extraction ratio (10 g soil in 50 mL water):

$$Cl^- \text{ (mg/kg)} = Cl^- \text{ (mg/L)} \times [\text{Volume of extract (L)} / \text{Weight of dry soil (kg)}]$$

$$Cl^- \text{ (mg/kg)} = 60.265 \times (0.05 / 0.01)$$

$$= 301.325 \text{ mg/kg}$$

Result

Result: The available chloride (Cl^-) content of the soil sample was found to be 301.325 mg/kg (301.325 ppm), indicating a low-to-moderate level of chloride with no apparent chloride toxicity. This level is not normally harmful to most crops.

Discussion

The two replicate titrations of the soil extract gave identical titre values (3.5 mL each), indicating good reproducibility of the titration end point. Chloride toxicity in sensitive crops is generally associated with soil chloride levels well above 350–700 mg/kg (thresholds vary considerably with crop, soil type, and irrigation water quality), so the measured value of about 301 mg/kg is unlikely to pose a toxicity risk under normal conditions, though it should be monitored where chloride-sensitive crops or saline irrigation water are involved. The main sources of error in Mohr's method are over-titration past the true end point (the pale brick-red colour can be subtle against the yellow background of the chromate indicator) and interference from other anions that may co-precipitate with silver; these can be minimised by titrating slowly with continuous swirling and comparing the end-point colour against the blank.

Experiment 8: Determination of Potassium (K^+) in Soil by Flame Photometer

Objective

To determine the amount of available potassium (K^+) present in a soil sample using a flame photometer.

Principle

When a solution containing potassium ions is introduced into the flame of a flame photometer, the potassium atoms are excited by the heat of the flame. As the excited atoms return to their ground state, they emit light of a characteristic wavelength — for potassium, at approximately 766 nm. The intensity of the emitted light is proportional to the concentration of potassium in the solution within the working range of the instrument. The potassium concentration of the soil extract is determined by comparing its flame-emission intensity with that of standard potassium solutions.

Reagents

- 1 N ammonium acetate (NH_4OAc) solution, pH ≈ 7.0 (used to extract exchangeable/available potassium)
- Standard potassium solution prepared from KCl (1.907 g dried KCl dissolved and made up to 1 L ≈ 1000 mg K/L stock); working standards of 0, 5, 10, 15 and 20 mg K/L prepared from the stock
- Distilled water

Procedure

A. Preparation of soil extract

1. 5 g of air-dried soil was weighed into a clean conical flask, and 25 mL of 1 N NH_4OAc solution was added.
2. The mixture was shaken thoroughly for about 30 minutes, then filtered through filter paper to obtain the clear soil extract.

B. Flame photometer measurement

1. The flame photometer was switched on and allowed to stabilise, and the K filter/wavelength (≈ 766 nm) was selected.
2. Distilled water was aspirated to zero the instrument.
3. The potassium standard solutions (5, 10, 15, 20 mg/L) and a blank were aspirated in turn, and their readings were recorded.
4. The soil extract was aspirated and its reading was recorded.
5. The potassium concentration of the soil extract was determined from the standard calibration curve after applying blank correction.

Observation

No.	Solution	K concentration (mg/L)	Observed reading	Corrected reading
1	Standard 1	5	0.4	0.1
2	Standard 2	10	1.0	0.7
3	Standard 3	15	1.4	1.1
4	Standard 4	20	2.1	1.8
5	Blank	0	0.3	0.0
6	Soil extract	Unknown	2.1	1.8

Corrected reading = Observed reading − Blank reading (0.3). The observed (raw) readings still contain the instrument's background signal and are therefore not used for quantification directly; only after this constant blank is subtracted does the reading correspond purely to the potassium signal. Accordingly, the calibration curve (Fig. 8.1) is constructed solely from the blank-corrected readings of the standards, and the corrected soil-extract reading is located on this same corrected scale so that it is directly comparable with the standards.

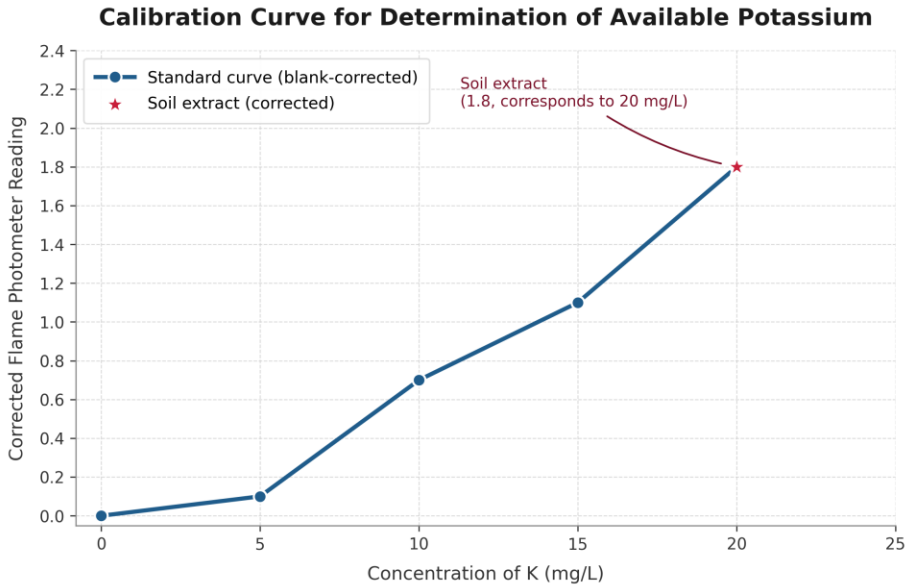


Fig. 8.1: Blank-corrected calibration curve for the flame-photometric determination of available potassium, with the soil-extract point marked

Blank Correction

Corrected soil reading = Soil reading − Blank reading = 2.1 − 0.3 = 1.8

The corrected soil-extract reading (1.8) coincides exactly with the corrected reading of the 20 mg/L potassium standard (2.1 − 0.3 = 1.8), so the soil-extract point falls precisely on the calibration curve at that standard. Hence, C = 20 mg K/L.

Calculation

Given: weight of soil, W = 5 g; volume of extracting solution, V = 25 mL; K concentration in soil extract, C = 20 mg/L; dilution factor, D = 1.

$$\textit{Available K (mg/kg)} = (C \times V \times D) / W$$

$$\textit{Available K} = (20 \times 25 \times 1) / 5 = 100 \text{ mg K/kg soil}$$

Result

Result: The available potassium content of the given soil sample was found to be 100 mg K/kg soil.

Discussion

Using blank-corrected readings for both the standards and the soil extract (rather than mixing raw and corrected values) ensures that the calibration curve and the sample point are on the same basis; in this data set the corrected soil reading happens to coincide exactly with the corrected 20 mg/L standard, giving an unambiguous concentration estimate without needing to interpolate between points. Available soil potassium of 100 mg/kg falls in the low-to-medium range commonly used to classify agricultural soils (values below roughly 100 mg/kg are typically considered low, 100–200 mg/kg medium, and above 200 mg/kg high, though exact thresholds vary by crop and local guidelines), suggesting that potassium fertilization should be considered for potassium-demanding crops. Possible sources of error include incomplete extraction of exchangeable K during shaking, contamination of glassware with trace potassium, and matrix/ionization interference in the flame photometer, all of which can be minimized by using matrix-matched standards and running replicate aspirations.